

No. My SQL Questions

What is MySQL

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What is Sql?

What is the difference between SQL and MySQL?

SQL Comments?

||[Difference between CHAR vs VARCHAR]|

What is MySQL?

- MySQL is an open-source relational **database management system** (RDBMS) that uses SQL to store, manage, and retrieve data.
- It's commonly used for managing data in web applications and is known for its performance and ease of use.

What is Sql?

- SQL stands for **structure query language**.
- SQL (Structured Query Language) is a standard language used to **create, read, update, and delete** data in relational databases. It is also used to create and modify database structures such as tables.
- SQL language hai, database nahi.

What is the difference between SQL and MySQL?

- SQL is a standard language used to communicate with relational databases, whereas MySQL is an RDBMS that uses SQL to store, manage, and retrieve data.
- SQL → Language
- MySQL → RDBMS / Database Management System

Difference between CHAR vs VARCHAR

- Both of these data types are used for characters.
- CHAR is a **fixed-length data type**, whereas VARCHAR is a **variable-length data type**.
- CHAR is suitable for **fixed-size values**, while VARCHAR is suitable for **values whose length can vary**.
- char occupies all the space and if space is remaining, then it fill all the blank space with "space". But in case of varchar, It takes only the required length & release remaining.

```
Char -> 10      | R | A | M | space | space | sapce | space | space | space |
space |
Varchar -> 10   | R | A | M |   |   |   |   |   |   |
| R | A | M |
```

- varchar is better than Char in term of space.
- char perform is better than varchar.
- Char max 256 characters, varchar 65535 characters.

```
-- CHAR Example
CREATE TABLE users (
    country_code CHAR(2)
);

INSERT INTO users (country_code)
VALUES ('IN'), ('US'), ('UK');

-- Yahan CHAR(2) suitable hai kyunki har country code ki length fixed 2 characters
hai.

--
-- varchar example
-- jaise name
CREATE TABLE users (
    name VARCHAR(50)
);
```

SQL Comments?

- There are typically two main types of SQL comments:

1. **Single-line Comments:** Started with two dashes (--)

```
-- This is a single-line comment  
SELECT * FROM employees; -- Retrieve all employee records
```

2. **Multi-line Comments:** Started with /* and ended with */

```
/* This is a  
   multi-line comment  
   explaining the query */  
SELECT id, name, email FROM employees;
```